Ex: 4

IMPLEMENTATION OF CISCO PACKET TRACER

AIM:

* To implement subnetting in cisco packet tracer simulator

INTRODUCTION:

*) Subnetting is the process of dividing a large IP network into smaller networks called subnets

PROCEDURE:

1 Create Network Topology:

* Open cisco Packet Tracer and create a blank workspace.
* Drag and drop components: 2 Routers (R1, R2), 2 Switches (S1, S2) and PC3 (3 Per Subnet)
* Connect devices using copper straight-through cables (Router to Switch, Switch to PCs)

2) Subnet schema

Subnet Range	Router interface	PC IP RANGE
192.168.1.0/27	R1(Gi0/0): 192.168.1.1	192.168.1.11 - 192.168.1.15
192.168.2.0/27	R1(Gi0/1): 192.168.2.1	192.168.2.11 - 192.168.2.15
192.168.3.0/27	R2(Fa0/0): 192.168.3.1	192.168.3.1 - 192.168.3.1

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*) Routes configuration:

enable
configure terminal
interface GigabitEthernet 0/0
ip address 192.168.11.255.255.255.224
no shutdown
Exit

*) Switch Port configuration:

enable
configure terminal
interface FastEthernet 0/1
Switchport mode access
exit

* PC IP Assignment

Open PC Desktop -> IP configuration - Assign static IP, Subnet Mask, default gateway.

* Testing network connectivity

Open Cmd on PC1 and execute Ping <destination IP> to verify intra-subnet and inter subnet communication.

RESULT:

The subnetted network was successfully created and configured in cisco Packet tracer using the subnet mask.